Week 1:

1.1 Preferences

unrelated to price/income

Consumer Preferences -> What do you want?

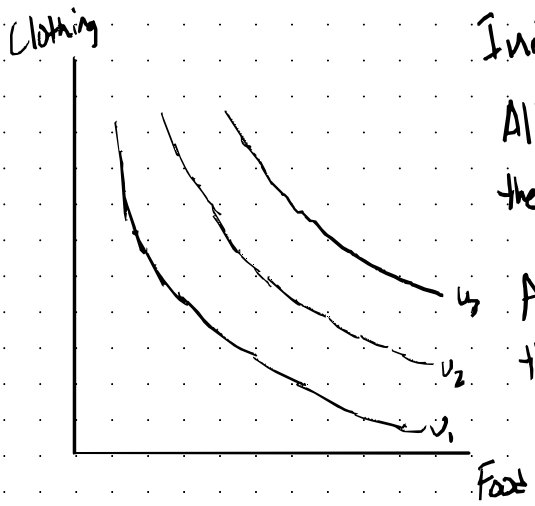
Budget Constraint -> What can you afford

Consumer Choice -> What will you do?

Market bundle: group of quantities of goods

Assumptions about Preferences:

1. Can rank and compare all baskets: >, <, =
2. Preferences are transitive: if A>B, B>C => A>C
3. More is always better



Indifference Curve:

All combinations on a curve have the same satisfaction level

A map of IC's shows all the different baskets

NE curves > SW curves
: more is better

Indifference Maps:

Can't be upward sloping

Can't be thick

Can't intersect with each other

Utility & Utility Functions:

Scores to represent satisfaction

ex: if $U(F,C) = 4FC^2$
what function for (1,9)

$U(1,9) = 4(1)(9)^2 = 324$

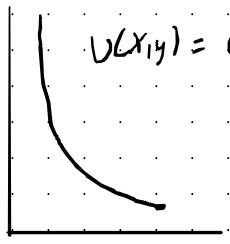
$4FC^2 = 324$

$FC^2 = 81$

$C^2 = 81/F$

$C = 9/F^{.5}$

Week 2: 3.1 Preferences



$U(x,y) = Cx^a y^b$

Marginal Rate of Substitution:

$MRS = \left| \frac{\Delta y}{\Delta x} \right| = -\frac{y}{x}$

The MRS_{xy} = max amt. of good Y a person gives up for 1 unit x

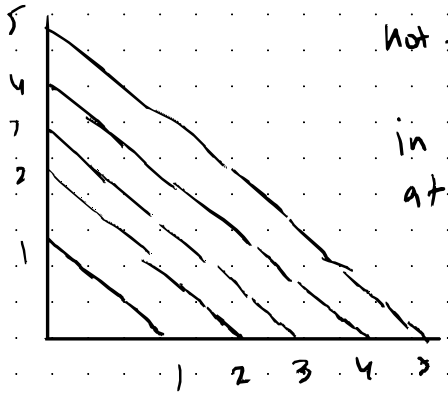
As F increases, MRS_{FC} decreases

↳ satisfies the assumption of diminishing MRS

Perfect Substitutes:

total product determines utility, not the quantities themselves

in a curve, you don't give up y for at a constant rate

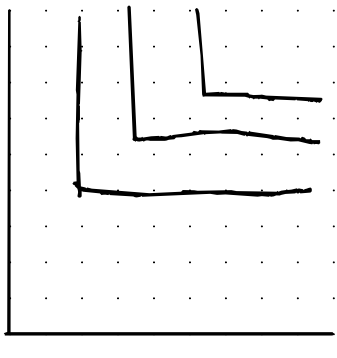


→ violates assumption diminishing

$U(x,y) = ax + by$

Perfect Complements:

→ slope is zero or infinity



only want to consume 2 goods in certain proportions

→ Violates assumption diminishing MRS

MU: +U per +x or +y: the rate at which utility increase as consumption of some good increases

$MU_x = \frac{\Delta U(x,y)}{\Delta x}$

$MU_y = \frac{\Delta U(x,y)}{\Delta y}$

→ are partial derivatives of $U(x,y)$

$MRS = \frac{MU_x}{MU_y}$

Week 2: 3.2 Budget Constraints

$I=80,\ P_F=1,\ P_C=2$

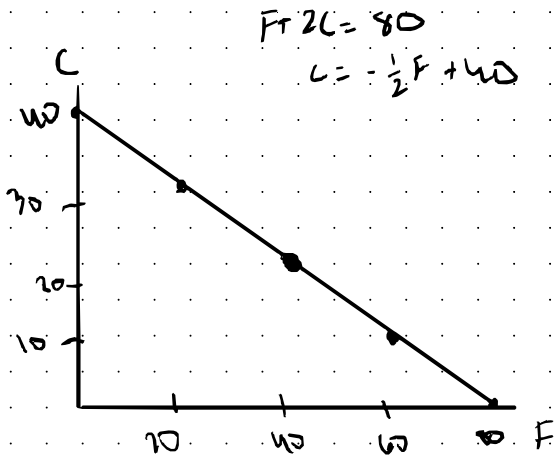
$F+2C=80$

$C=-\frac{1}{2}F+40$

$$P_x \cdot x + P_y \cdot y = I$$

$$\downarrow$$

$$y = -\frac{P_x}{P_y}\cdot x + \frac{I}{P_y}$$



Budget line slope: $\frac{-P_x}{P_y}$

y-intercept: I/P_y

x-intercept: I/P_x

*income change = budget line shifts parallel to original line

* price change = budget line rotates about an intercept

Utility Maximization Problem:

Default case: Diminishing MRS, smooth

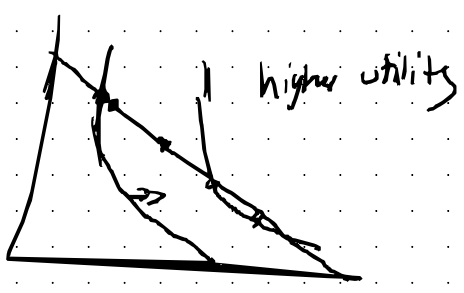
$$P_x \cdot x + P_y \cdot y = I$$

$$MRS_{xy} = \frac{P_x}{P_y}$$

$$\frac{MU_x}{MU_y} = \frac{P_x}{P_y}$$

*If $MRS_{xy} > \frac{P_x}{P_y}$ = slope not same

increase use of x, to move to higher IC on same BC



ex: $U(x,y)=x^{.5}y^{.5}$, $MRS_{xy}=\frac{y}{x}$ $I=100$ $P_x=10$ $P_y=2$

$10x+2y=100$ $MRS=\frac{P_x}{P_y} \Rightarrow \frac{y}{x}=\frac{10}{2}=5$

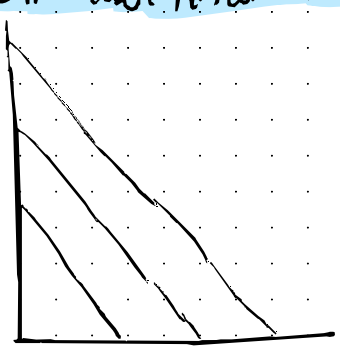
$\Rightarrow y=5x$

$10x+2(5x)=100$

$20x=100$

$\underline{x=5} \Rightarrow y=25$

Other case: Perfect Substitutes



$P_x=1$ $P_y=2$ $I=10$

$U(x,y)=5x+4y$

$MRS \Rightarrow 1=5x+4y$

$y=-\frac{5}{4}x+\frac{1}{4}$

$MRS=\frac{5}{4}$ $\frac{P_x}{P_y}=\frac{1}{2}$

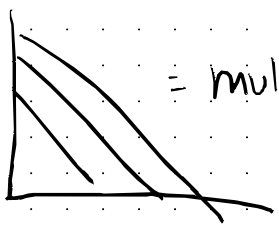
$>$

preference for x, spend all on x

but in example:

$U(x,y)=5x+4y$ $P_x=10$ $P_y=8$ $I=80$

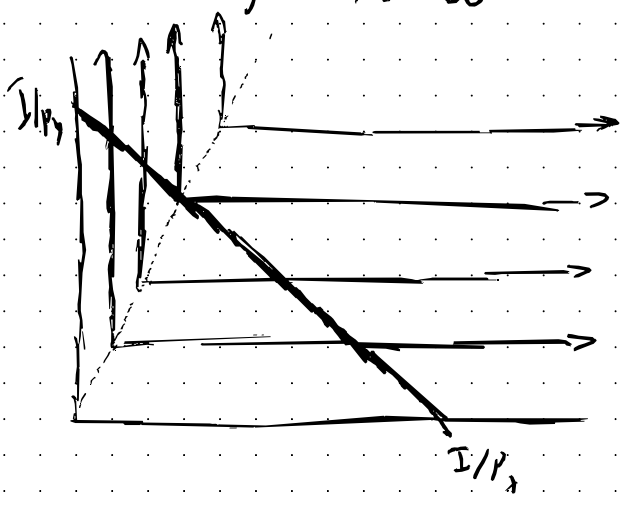
$MRS=\frac{5}{4}=\frac{P_x}{P_y}=\frac{5}{4}$



= multiple optimal baskets

Other case: Perfect Complements

$P_x=2$, $P_y=4$, $I=60$ kinks on $y=2x$



utility max at a kink

(x^*, y^*)

satisfies: $y=2x$

$2x+4y=60$

$(x,y)=(6,12)$